

**"ANOMALOUS PRICE BEHAVIOR AROUND
REPURCHASE TENDER OFFERS"**

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ABSTRACT

This paper reports anomalous price behavior around repurchase tender offers. First, buying shares before the expiration date of a repurchase tender offer and tendering to the firm produces excess returns of more than 9% per week, on average. Second, it is found that repurchasing companies experience economically and statistically significant excess returns in the two years after the repurchase. The upward price drift is mainly generated by the behavior of very small firms in the sample.

INTRODUCTION

In this paper we test for market efficiency by analyzing the performance of two trading rules around repurchase tender offers.

Stock repurchases have been analyzed extensively by Dann (1981), Masulis (1980), Rosenfeld (1982) and Vermaelen (1981, 1984). These studies are event studies which focus on explaining the abnormal price increase around repurchase tender offer announcements. Dann and Vermaelen conclude that information signalling explains, more than any alternative hypothesis, the abnormal returns after the announcement. However, stock repurchase tender offers also provide an interesting setting to examine market efficiency, for two reasons.

First, a repurchase tender offer creates a lot of price uncertainty. In order to price securities properly during the tender offer period, investors have to estimate (1) the fraction of shares tendered by the other shareholders, (2) the subsequent repurchase decisions by the management when the offer is oversubscribed and (3) the market price after the expiration of the offer. Hence, a first trading rule tests whether on average stock prices during the offer period preclude arbitrage opportunities. Specifically, we test whether it is possible to make abnormal returns by buying shares before the expiration date and tendering to the company. The trading rule mimics the behavior of risk arbitrageurs who typically buy and tender during the offer period. This is at least the view of practioners such as Weinstein (1984) who states that "the investor must keep accurate statistics of the daily trading volume from the time of the announcement of the self-tender until the early pro rata date or registration date of the offer as an indication of how many shares are likely to be tendered. Most of the stock will have been bought by arbitrageurs with the intention of tendering ...".

Larcker and Lys (1987) conclude that risk arbitrageurs earn substantial returns on their trading activities around mergers, intra-firm tender offers and voluntary liquidations. They argue that risk arbitrageurs are able to generate private information regarding the success of such corporate reorganizations. Alternatively one could argue that these excess returns are a fair compensation for the services that arbitrageurs in takeover bids are said to provide (Jensen (1986b)): (1) helping

to evaluate alternative offers (2) providing risk-bearing services for investors who rather sell than bear the uncertainty surrounding a restructuring and (3) resolve the free-rider problems of small, diffuse target shareholders who cannot organize to negotiate directly with the bidder. In the case of repurchase tender offers, only the second service is relevant. Hence, considering the simplicity of the trading rule and the relatively small number of services the arbs provide during repurchase tender offers, we do not expect to earn large returns from just buying and tendering.

A second reason why a repurchase tender offer provides an interesting setting for examining market efficiency follows from the results reported in previous research. It is found that, on average, tendering shareholders receive premiums of 23%, while non-tendering shareholders obtain returns of only 14%. As the non-tendering shareholders include the management of the company, (see Vermaelen (1981)), insiders apparently “give away” part of the firm to outsiders (at least, if the repurchase reveals information which would have become known anyway through earnings reports, etc.). One reason for doing this may be that, at least for the management, there are offsetting signalling benefits to compensate for the expropriation of their personal holdings. For example, the repurchase may be a way to fight a potential takeover bid so that managers can keep on consuming perks out of firm resources. (Bagwell (1988), Bagnoli, Gordon and Lippman (1987), Vermaelen (1984)). Or, insiders may buy shares before the announcement and sell them afterwards (Choi (1987)). An alternative hypothesis may well be that the market does not fully realize (around the announcement) how costly the repurchase is to the non-tendering management and therefore underestimates the true value of the shares. This implies that buying shares after a repurchase tender offer should be a good investment, especially in cases where the market apparently does not believe that the signal is very credible and at the same time the (by the market perceived) wealth loss to the insiders is very large. Specifically, in cases where the managers own a substantial stake in the company, and where the firm buys back a large fraction of its own stock at large premiums and, at the same time, the market does not react very positively around the announcement, one may expect to earn positive excess returns after the repurchase. Hence, the second trading rule involves buying and holding shares after the expiration date of a repurchase tender offer. Interestingly, in a recent Fortune magazine study, Loomis (1985) argues that buying stock after a repurchase generated excess returns (above the S & P 500) of 8.5% per year in

the period 1974 though 1983.

This paper is organized as follows. In section two we describe the data base. In section three we test the first trading strategy: buying stocks before the expiration date and tendering to the firm, whenever the market price is significantly below the tender price. The simple strategy generates economically and statistically significant abnormal returns of 9% on average. In section four we examine whether the excess returns result from the peculiar repurchase behavior of management in over-subscribed offers. In section five we test a second trading strategy: buying shares after the expiration of the offer. It is found that we can beat the value-weighted index by 12% per year in the two years after the repurchase. After adjusting for size and beta the “excess” returns fall by more than half but are still significantly positive. Further investigation reveals that these excess returns are mainly created by very small firms. Section six summarizes our results and suggests some topics for further research.

2 Data

The data base consists of the announcement and expiration dates, the terms (fraction sought, tender price) and outcome (fraction tendered, fraction purchased) of all 258 repurchase tender offers which occurred between 1962 and 1986 by firms traded on the NYSE, AMEX and OTC. The first set of data covers 131 observations between 1962 and 1977 and is adopted from Vermaelen (1981). Data for 1978 and 1979 were taken from the Wall Street Journal Index which started summarizing repurchase announcements in 1978. From 1980 on we had access to all the 13-e4 filings and the filing date. We used the filing date to track the repurchase in the Wall Street Journal Index. If no announcement was found, a questionnaire was sent to the corporation. For the firms that did not reply, we searched the Wall Street Journal Index for an announcement (advertisement) of the event. Repurchases intended to reduce small shareholder servicing costs, were deleted from the sample. Data on stock prices and returns were taken from the daily return and master file from the Center for Research in Security Prices of the University of Chicago, for NYSE and AMEX stocks. Data on the OTC stocks were obtained from the Standard and Poor's Daily Stock Price Record. Data on the daily trading volume for NYSE and AMEX stocks were taken from the Cornell University Price and Volume

File (CUPV) which contains data on the daily number of shares traded, opening, closing, high and low prices (for a more extensive description, see Lakonishok and Smidt (1984)). Data on trading volume for OTC companies were obtained from the Standard and Poor's Daily Price Record.

Table 1 provides a number of descriptive statistics for 221 observations for which data on the fraction of shares sought, fraction tendered and fraction purchased were available. On average, in the total sample period, firms offered to buy back 17.06% of their shares at a premium of 21.79% above the market price 5 days before the repurchase announcement. On average, 16.41% of the shares outstanding were purchased, but a much larger fraction (87%) of the shares tendered was purchased by the company.

We also computed the cumulative average abnormal return ¹ to the non-tendering shareholders from 5 days before the announcement until 5 days after the expiration date (*CAR*) and the weighted average abnormal return to the tendering and non-tendering shareholders (*TOTALR*). On average, non-tendering shareholders earn abnormal returns of 12.54%, which is significantly smaller than the 21.79% premium that the tendering shareholders receive. Because, on average, 16.41% of the outstanding shares are repurchased, the repurchase increases the total value of the firm by 14.29%.

To test for the stability of these figures, we divided our sample in two subperiods: (1) 1962-1979, which corresponds approximately to the period examined in past research and (2) the more recent experience: 1980-1986. Table 1 shows that in the more recent time period, returns to tendering and non-tendering shareholders became smaller. We elaborate on these results *infra*.

3 Trading rules around the expiration date

3.1 Methodology

¹Returns were adjusted for the CRSP equally weighted market index.

In a repurchase tender offer, firms offer to buy back a fraction of their shares at a tender price, P_T , before a specific date (the expiration date).

A first trading rule consists of buying shares before the original expiration date ² and tendering to the firm. In order to compute the gains from such a strategy, it is important to keep in mind the rules governing the repurchase tender mechanism.

If the offer is undersubscribed (ie. the fraction of shares tendered, F_T , is less than the fraction of shares sought) the firm will repurchase all shares tendered. This is true, even when the offer is extended and becomes oversubscribed later on. If the offer is oversubscribed, the firm will either buy back all the shares tendered, or allocate pro-rata so that each shareholder can sell a fraction F_p/F_T of his shares to the company (where F_p is the fraction purchased).

Thus, the profit from buying a stock at a price P_B during the tender period and tendering it to the firm is equal to $P_T - P_B$ if the offer is undersubscribed at the first expiration date. If the offer is oversubscribed at the first expiration date, the profit from such a strategy is equal to

$$P_T \frac{F_p}{F_T} + (1 - \frac{F_p}{F_T}) P_E - P_B \quad (1)$$

where P_E is the price after the expiration day. In cases where $P_B > 0.97 P_T$ no trading was undertaken, in order to allow for transaction costs.

To test for the sensitivity of the results to the timing of purchases and sales, different trading strategies were tried out. The first trading rule assumes buying and tendering on the expiration day and, if the offer is oversubscribed, selling the non-repurchased shares two days after the expiration day. In the second trading rule we assume that we purchase on the day before the expiration day and sell two days after. Finally, the third trading rule assumes that

²An offer may be extended so that several expiration dates exist. In total 99 of the 258 offers were undersubscribed at the initial expiration day. 43 were extended and 11 of these became subsequently oversubscribed.

we buy on the day before the expiration day and sell four days after. In each case we examine the total observation period (1982-1986), a first subperiod (1962-1979) and a second subperiod (1980-1986).

Vermaelen (1981) presents a similar trading rule, but looks at the total sample. In this paper we only analyze a subsample where the market price is at least 3% below the tender price. He also uses ex-post information, because his trading rule is tested round the final expiration date. Because approximately half of the 99 undersubscribed offers are extended, the final expiration date is unknown ex ante. Moreover, when an undersubscribed offer is extended and eventually becomes oversubscribed, the firm will purchase all shares tendered before the earlier expiration date(s) prior to allocating pro rata. Hence, for these offers, equation (1) is not valid ³.

3.2 Results

The results of the trading strategy are shown in table 2. The abnormal returns presented are the returns from the trading strategy adjusted for the market using the CRSP equally-weighted index. Because the holding period is short, the results are not sensitive to the assumption made about the process that generates returns (see Brown and Warner (1985)). Part A of the table (which covers the total sample period) shows that in 109 out of the 258 cases, the market price is more than 3% below the tender price before the expiration date. The trading rules generate a rather impressive abnormal return of more than 9% on average, with t-statistics larger than 7.5. The median value is smaller (=5.5%) but, in the worst case, (trading rule number 3) 93% of the abnormal returns are positive. There seems to be no large difference between the different trading rules. The worst possible outcome is a loss of 7.1%.

Panels B and C show similar results, but now for two subperiods. It seems that, on average, the excess return has increased in the eighties: approximately 11% vs. 8% in the sixties and seventies. However, this difference is

³However, those cases are extremely rare: of the 43 extended offers, only 11 became subsequently oversubscribed.

not statistically significant. Because the variability of returns has increased in recent years, t-statistics are higher in the first period: approximately 7 vs. 4.5 in the second subperiod. Note also that median excess returns remain approximately stable at 5.5% and that in at least 25% of the cases, for all subperiods and trading rules, excess returns larger than 8.5% were obtained. The persistence of abnormal returns is striking, considering that in the second subperiod, for two out of the three trading rules, the excess returns are always positive.

In order to test whether the performance of the trading rule depends crucially on the (arbitrary) decision to consider only cases where the pre-expiration price is at least 3% below the tender price, the trading rule was repeated with a 5% cut-off rate.

The results in table 3 show that, while the number of observations falls by approximately 30%, the average and median excess returns from the trading strategy are larger than in table 1, for all sub-periods. To test for the generality this result, the regression

$$XRET_i = \alpha + \beta \frac{P_T - P_B}{P_B} + e_i \quad (2)$$

was run, where $XRET_i$ is the excess return from the trading strategy. All 193 cases in which the pre-expiration price was below the tender price are included in the sample.

Table 4 shows the regression coefficients and R^2 for the various trading rules. The degree of underpricing (below the tender price) explains more than 75% of the variance of the trading returns and the regression coefficients are highly significant ($t > 23$). The regression coefficient of .85 implies that buying shares before the expiration date whenever the market price is x percent below the tender price, produces abnormal returns of .85 x percent.

Figure 1 plots the dependent variables against the independent variable (for the first trading rule) and shows that the results are heavily influenced by four major outliers. Hence, a non-parametric Spearman-rank correlation

test was performed. The rank correlation coefficients for the three trading rules are equal to 0.687, 0.667 and 0.685 respectively, and are all significantly positive. Thus, it is concluded that the performance of the trading rule can be significantly enhanced by focusing on cases where the pre-expiration price is significantly below the tender price.

3.3 Adjustment for risk

The results presented assume that the risk of the trading rule is similar to the risk of an investment in the equally-weighted market index. However, to the extent the firm buys back the shares tendered the strategy is essentially riskless. If the firm returns some of the shares tendered the investor has to bear market risk on the non-repurchased shares. Thus, it is clear that the average investment will be less risky than an average investment in the stock market, so that the abnormal returns are underestimated.

Note that the trading rule assumes that we can actually sell the shares immediately after the expiration date. When an offer is oversubscribed the non-repurchased shares are returned up to two weeks after the offer expiration date. However, the investor can cover this additional risk by shortselling the non-repurchased shares after he finds out about the pro-rata decision (which occurs, in general, the day after the expiration date).

3.4 Transaction costs

An investor who buys shares and tenders incurs a one-way transaction cost on all repurchased shares. When selling the remaining shares after expiration of the offer, an additional transaction cost has to be paid. Hence, transaction costs will always be less than the cost of one round-trip transaction. Moreover, as shown in table 4, investors who face high transaction costs can always focus on the “deep discount” offers.

3.5 Liquidity

One explanation for the persistent underpricing may be the fact that, because of thin trading, risk arbitrageurs can't trade the anomaly away. In order to test this hypothesis, a measure of "normal" trading volume was estimated as the average number of shares traded per day in the period from 50 days before the announcement until 25 days before the announcement. Data were taken from the Cornell University Price and Volume file (CUPV). Because the file only covers the period from January 1, 1970 until December 31, 1981, the tests are based on a smaller number of observations. The ratio of trading volume relative to estimates of "normal" volume was computed for each firm in the sample, and averaged across firms, for two event dates: (1) from 10 days before the announcement day until 10 days after and (2) from 10 days before the first expiration date until 10 days after. We excluded the observations where there was more than one expiration day (i.e. extended offers). In each case we examine the total observation period and two subperiods.

Table 5A shows the results around the repurchase announcement. Only during the three days prior to the announcement is trading volume significantly higher than normal in all three sample periods. Jarrell and Poulsen (1988) report that trading volume before intra-firm tender offers is at least double what it was over a control period as early as 10 days prior to the announcement date. Thus, repurchase tender offers are more unexpected events than takeover bids and are not preceded by significant insider buying ⁴. Table 5A shows that in each of the 10 days after the announcement the abnormal trading volume is significantly positive and at least 2.5 times what it was over the control period.

The results in table 5B (which centers around the expiration date) show that, on average, in both periods the trading volume was at least twice as large as normal in each of the 10 days prior to and including the expiration day. The largest abnormal volumes are observed early in the tender period when

⁴Choi (1987) examines the Official Summary of Insider Transactions around repurchase tender offers announcements and does not report any abnormal insider transaction behavior in the 12 months preceding the repurchase.

the number of shares traded is sometimes more than 500% above the normal level. After the expiration day, the trading volume is often smaller than normal. The results for the median and “percentage larger than 1” statistics also show a pronounced difference in trading behavior before and after the expiration date. Comparing the results for different subperiods, we can conclude that the typical trading volume pattern is stable over time.

It is concluded that there is clearly not a lack of market liquidity during the offer period. The abnormally large trading volume may reflect the willingness of investors to sell their shares at a discount to arbitrageurs in order to avoid the uncertainty surrounding the event. On the other hand, if the large trading volume reflects the activity of professorial arbitrageurs who buy and tender (as suggested by Weinstein (1984)), it is puzzling why they have not arbitrated the abnormal profits away. The market apparently underestimates F_p/F_T and/or P_E . With respect to the first variable, it is interesting to note that, on average, a very large percentage (87%) of all shares tendered were repurchased by the company.

3.6 Cash settlement vs. 5-day settlement

An investor who buys shares through a “normal” transaction has to wait 5 business days before he or she receives the stock. Hence, buying and tendering 1 day before the expiration date is not feasible. However, by following a special cash-settlement procedure it is possible to pay and receive the shares on the same day. Discussions with brokers revealed that this special procedure implies larger transaction costs for relatively small trades (i.e., a small investor can’t use discount brokers). Hence, for an ordinary individual investor, the trading rules described above may not be practically feasible. Thus, we repeated our trading strategy, but now assuming that the stock was purchased 6 business days before the expiration day and the non-repurchased shares are sold 12 business days after the expiration day. The long post-expiration period reflects the fact that a small investor may not be able to cover his or her risk by shortselling immediately after the expiration of the offer. Indeed, borrowing shares may be a problem in highly oversubscribed offers.

The results in table 6 show the excess returns from the trading strategy, for two cut-off rates: 3% (Panel A) and 5% (Panel B). The average and median returns are always smaller and less significant than their corresponding values in tables 2 and 3, but still significantly positive. On average, excess returns are always larger than 5.5%, for both cut-off rates and in all subperiods, and in approximately 90% of the cases, excess returns are positive.

In order to test for the relationship between cut-off rates and trading excess returns, the regression

$$XRET_i = \alpha + \beta \frac{P_T - P_B}{P_B} + e_i \quad (3)$$

was run. The regression results are consistent with the ones reported in table 4: $\alpha = -0.02(t = -3.62)$, $\beta = 0.86(t = 18.85)$ and $\bar{R}^2 = 0.65$. The Spearman rank correlation coefficient between the dependent and independent variable is again positive ($= 0.46$) and significant at the .1% significance level.

Hence, we conclude that the small investor should trade-off the larger transaction cost of a cash settlement deal (and shortsale) against a lower expected return (of around three percent, on average) and higher risk from holding the non-repurchased stock during a longer time period.

3.7 Conclusion

The results show that (1) investors can make positive excess returns by purchasing shares which are trading at a discount from the tender price and tendering them to the firm. (2) The profits from the strategy are a positive function of the size of the discount. (3) Investors who buy at least 6 days before the expiration date and sell the (non-repurchased) stock 12 days after expiration realize, on average, smaller returns than investors who can buy and sell very close to the expiration date. On the other hand they avoid the (for small investors) higher transaction cost of a cash settlement deal and a shortsale. (4) The returns realized in the trading rule seem too large to

be explained as a compensation for risk-bearing, especially considering that the strategy is considerably less risky than an average investment in common stock. (5) The returns can also not be explained as a fair compensation for costly information acquisition (as argued by Larcker and Lys (1987)) because the trading rule requires no sophisticated analysis.

4 Trading profits and repurchase decisions

The reason why risk arbitrageurs can't compete the anomaly away may have something to do with the fact that, at least in oversubscribed offers, the firm's management ultimately determines the magnitude of the trading returns. Recall that, when an offer is oversubscribed, the managers can buy back the originally stated target number of shares and allocate pro-rata, or purchase more. If managers systematically buy all shares tendered when the stock is selling at a (large) discount from the tender price prior to expiration, but allocate pro-rata otherwise, the arbitrageur faces a dilemma: if he drives up the price to the tender price, the management will allocate pro-rata; but, if the price after the expiration date is systematically below the tender price, the arbitrageur will systematically lose on all the non-repurchased shares. However, if the price prior to expiration remains below the tender price arbitrage profits will exist if (as assumed) the management buys all shares tendered! In other words, because in oversubscribed offers the management ultimately decides on the arbitrage profits, equilibrium no-arbitrage prices may not exist.

Why would management behave in such a way? If the purpose of the offer is to increase stock prices in order to correct mispricing of their securities and the market does not react enthusiastically enough to the signal (i.e. $P_B < P_T$), the management can strengthen the signal ex-post by purchasing more than originally intended⁵.

This explanation subsumes a number of testable hypotheses. First, the abnormal

⁵ A classic example of this behavior is the repurchase offer made by Teledyne on September 15, 1972 for 1,000,000 shares at a price of \$20 per share. On the announcement date the stock price rose from \$16.50 to \$17 7/8 and remained relatively constant until the expiration date when it fell to \$17.50. 8,000,000 shares were tendered and on October 18 Teledyne decided to borrow \$140 million to buy back all shares tendered. On that day, the stock rose to \$20.40. For more details see the HBS case "Synerdyne".

returns are mainly realized in oversubscribed offers. This hypothesis is easily confirmed by the fact that from all the offers where the market price 1 day prior to expiration is at least 3% below the tender price, 78% are oversubscribed.

Second, the reasoning implies that in a typical rationed oversubscribed offer the price after the expiration will fall. At the same time, in the case where the management purchases more than it has to, the market will react positively. Evidence consistent with this hypothesis is provided in table 7, where average and cumulative average daily abnormal returns are computed from 2 days before the expiration day until 6 days after. In the total sample (Panel A) stock prices decline significantly and permanently by 1.2% on the day after expiration. When the firm buys at least 20% shares more than it has to (Panel B) the negative return on day 1 is more than compensated so that, on average, stock prices increase from the day after the expiration date until 5 days after by 1.56%. Hence management can influence post-expiration stock prices by its repurchase decision.

Finally, the reasoning above assumes that management will be influenced by the price behavior prior to the expiration date: if the price prior to expiration is significantly below the tender price, it will buy relatively more than it has to, because this price behavior reflects the market's expectation that $P_E \ll P_T$. This hypothesis was tested by regressing, for all oversubscribed offers, the relative discount from the tender price (P_B/P_T) against F_P/F_T . The results in table 8 provide only mild support for the hypothesis that managerial decisions are taken in function of the pre-expiration price behavior. The coefficient on P_B/P_T has the expected sign but is not highly significant. Moreover, the variable only explains 1.26% of the cross-sectional variance of the repurchase decisions. In short, it is difficult to explain why risk arbitrageurs have not traded the anomaly away.

5 Trading rules after the expiration date

As shown in table 1, tender offer premiums exceed abnormal returns to non-tendering shareholders by almost 10 percent. Or, tendering shareholders are significantly better off than non-tendering shareholders. If the value change is an information signal, then the insider-managers of the corporation (who don't tender) incur a significant signalling cost (see Vermaelen (1981, 1984)). Specifically, if

the firm repurchases N_p shares at a price $P_T > P^*$, where P^* is the “true” stock price (known to insiders) then the $N_O - N_p$ non-tendering shareholders (where N_O is the number of shares outstanding before the repurchase) bear a cost of $(P_T - P^*) N_p$. If the managers-insiders own M shares of the firm and don’t tender, their share of the cost is $(P_T - P^*) N_p M / (N_O - N_p)$. The fact that the market price after the expiration P_E is significantly below the tender price P_T , implies that the perceived signalling cost $(P_T - P^m) N_p M / (N_O - N_p)$ is significant, where P^m is an estimate of the “true” price per share implied by the post-expiration price P_E . Indeed, per definition, $P^m = F_p P_T + (1 - F_p) P_E$, where F_p is the fraction of shares repurchased so that, if $P_E < P_T$, $P^m < P_T$. This allows us to propose two alternative hypotheses.

- H.1: The market is efficient (in the semi-strong form sense), or $P^m = P^*$. Managers incur the signalling cost because there are some offsetting benefits (eg. the possibility to deter a raider).
- H.2: The market systematically underestimates the value of P^* . Hence, the true signalling costs are smaller than the ones perceived by the market, or $P^* > P^m$.

If the second hypothesis holds, then buying stocks after the expiration date and holding them until P^* is revealed (eg. through earnings reports) should generate positive abnormal returns.

5.1 Computing abnormal returns

To study the long term price behavior of repurchasing firms after the announcement, we adopted the methodology proposed by Dimson and Marsh (1986) in their study of UK brokerage recommendations. The abnormal return for security i in day t is computed in four different ways: by subtracting an equally-weighted index and a value-weighted index, by adjusting for size and by adjusting for size and beta.

5.1.1 Market adjusted performance

As a first step in the analysis returns are adjusted for two market indices: the value-weighted CRSP index and the equally-weighted CRSP index.

Hence, we define the excess return for security i in day t as:

$$e_{it} = R_{it} - R_{vt}(\text{value weighted index})$$

or as

$$e_{it} = R_{it} - R_{et}(\text{equally weighted index})$$

5.1.2 Size-adjusted performance

Market values of equity of all firms on the NYSE and AMEX on the last day of each year starting in 1962 and ending in 1986 are computed. Stocks are ranked according to market value and 10 size-based portfolios are formed. The excess return in day t for security i is then computed as

$$e_{it} = R_{it} - R_{st} \quad (\text{size control})$$

where R_{st} is the return on a control portfolio, represented by the size decile which includes firms with approximately the same size as firm i .

5.1.3 Size and beta-adjusted performance

Betas of repurchasing firms and their control portfolios are computed by regressing stock returns against the value-weighted market index using 60 monthly observations before the repurchase month. The size and beta adjusted abnormal return is then computed as

$$e_{it} = R_{it} - R_{st} - (\hat{\beta}_i - \hat{\beta}_s)[R_{mt} - R_{ft}] \quad (\text{size and beta})$$

where R_{ft} is the Treasury Bill rate, and $\hat{\beta}_i$ and $\hat{\beta}_s$ are the beta coefficients of the repurchasing firm and its control portfolio, respectively.

Average abnormal returns and cumulative average abnormal returns are computed for all firms in the sample, from 2 days before the announcement until 480 days after. In the tables below we assume that 20 trading days is equivalent to a monthly return, so that the test period consists of 24 months after

the announcement date. Note that we use the daily CRSP file which includes AMEX and NYSE firms. In each case, we examine the total observation period (1962-1986), the first (1962-1979) and the second subperiod (1980-1986).

5.2 Results

The results using the 4 different benchmarks in the total period, the first and second subperiod are reported in Tables 9, 10 and 11 respectively. The results using the value-weighted index as a benchmark are reported in the first column of each table; equally-weighted index adjusted returns are reported in the second column; size-adjusted and size and beta-adjusted returns are shown in the last two columns.

5.2.1 Market Adjusted Returns

The announcement return in the total period is equal to 13.82%. The announcement has a more positive effect before 1980 (16.04%) than after 1980 (10.54%), in spite of the fact that, on average, the premiums and fraction purchase are not very different in both periods. In the first subperiod firms repurchase, on average, 16.7% of their shares at an average premium of 22.9%. The corresponding numbers in the second subperiod are 19% and 20% respectively.

A number of explanations can be proposed for this difference. First, because repurchases have become more popular in general, they are less unexpected than in the past. Second, managerial incentives to signal may actually have increased in recent periods. Vermaelen (1984) finds evidence that firms tend to make repurchase tender offers when there is a lot of takeover activity. This suggests that firms may want to signal in order to prevent being taken over at a too "low" stock price. Hence, the merger and takeover wave of the eighties may have encouraged managers to offer large premiums for the company's shares, even if these offers entailed a reduction in the value of their personal holdings. Third, the repurchase tender offer may have been a less convincing signal today than in the past, because the typical firm is larger and is

less controlled by insiders. Hence, lower insider holdings mean lower costs of “false” signalling and therefore lower credibility. Finally, the difference in announcement returns may reflect different motivations behind the repurchase activity: Jensen (1986a) argues that in recent years many firms have bought back shares in order to reduce the agency costs resulting from free cash-flows. Hence, insiders-managers were willing to bear a much larger cost to increase real cash flows than to increase perceived cash flows.

The first column of table 9 shows, for the total period, that from the end of the second month after the announcement until 24 months after the announcement, repurchasing stocks beat the value-weighted index with 23.11% (or 1.05% per month) ($t = 6.92$). Tables 10 and 11 show that this effect is more pronounced in the first period: 1.40% per month before 1980 ($t = 6.16$) vs. 0.47% per month in the second subperiod ($t = 1.97$). These results are comparable to the ones reported in the Fortune study (Loomis (1985)) that reports an excess return of 8.5% per year over the (value-weighted) S & P 500 index.

However, the second column of tables 9, 10 and 11 shows that the result depends very much on the market index used. When the returns are compared with an equally weighted index, the “abnormal” performance in the post expiration period becomes much smaller. From 2 months after the announcement until 24 months after the announcement, repurchasing firms earn abnormal returns of .56% per month in the total period ($t = 3.78$), .76% per month in the first period ($t = 3.66$) and .23% per month in the second period ($t = 1.00$).

The difference in results between the first and second column can be explained by the “size effect” (for a similar explanation of abnormal performance after broker recommendations, see Dimson and Marsh (1986)): a value weighted index will mainly reflect the performance of the largest firms. As small firms have historically outperformed large firms (see Banz (1981)), the post-announcement performance merely reflects the “small firm” effect. The difference in results between the two subperiods could be explained if the small firm effect was less pronounced after 1980 and/or in the second subpe-

riod the typical repurchasing firm was larger.

5.2.2 Size adjusted returns and size and beta adjusted returns

Table 12 shows that, although in the total period repurchasing firms are approximately evenly distributed across different size groups, an increase in firm size is observed in recent years. For example, in the the first period 4.5% of the repurchasing firms are in the largest size decile (group 10) while the corresponding number in the second subperiod is 18.9%.

The last two columns in table 9, 10, and 11 show the cumulative average abnormal returns adjusted for size and adjusted for size and beta. The two columns are almost identical so that we only discuss the results adjusted for size and beta.

The results show that, in the total period, stock prices increase significantly by .4% per month ($t = 2.70$) in the 22 months following the second month after the repurchase announcement. However, the results are only statistically significant in the first period when stock prices increase abnormally by 0.45% per month ($t = 2.23$). Although not statistically significant at conventional significance levels ($t = 1.41$) the excess return of .30% per month in the second subperiod is still an economically significant abnormal return.

5.3 Interpretation of the results

The results show that firms that offer to repurchase their own shares at a premium not only experience abnormal returns around the announcement, but also after the expiration date. The increase after the expiration date is such that firms (or better, non tendering shareholders), on average did not “overpay” when they offered premiums of 22.9%: the cumulative average abnormal return of 20.88% twenty-four months after the announcement date is not significantly different from the average premium. If one accepts “signalling” as the main incentive for the repurchase, this results implies that,

ex post, managers-insiders (who don't tender their shares) did not incur any significant signalling cost. It also means that the signal was not fully revealing around the announcement date: i.e. the market did not fully incorporate the good news revealed thru the repurchase.

The results also show that repurchases in recent years were made by larger firms that were to some extent "overpaying" for their own stock: the cumulative average abnormal return 24 months after the repurchase announcement is still more than 3% below the repurchase premium of 20%. In order to test whether the price behavior was also different in the period prior to the announcement date, the cumulative excess return (adjusted for beta and size) in the 40 months prior to the repurchase was computed.

In the first period (1962-1979) repurchasing firms earned negative excess returns of - 11.40% before the repurchase. In the second period (1980-1986) repurchasing firms earned significant positive abnormal returns of 15.10% ($t = 2.51$) prior to the repurchase. The results are consistent with the hypothesis that, after 1980, most repurchases were done by firms that were doing relatively well and may have accumulated large quantities of excess cash flows (as argued by Jensen (1986a)). Because managers were willing to incur costs to increase the real value of the firm (as opposed to the perceived value of the signalling hypothesis), the tender price is significantly higher than the (long run) post-expiration price. In the first subperiod (1962- 1979) managers were not willing to incur any costs because not the real value of the firm, but only the perceived value increased as a result of the repurchase.

5.4 Effect of firm size on size-adjusted abnormal returns

The previous results show that the market, at least in the period before 1980, has systematically under reacted to the information content of the repurchase. To test whether this conclusion depends on the size of the repurchasing firm, the previous analysis was replicated for five size-based portfolios. The cumulative average abnormal returns in the 40 month period prior to the repurchase (CAR (-40, -1)), in the announcement month (CAR (0,0)), and in the 23

months after the expiration ⁶ (CAR (2,24)) are shown in table 13.

The results show that the smaller the firm size, the larger the announcement returns (a result also observed by Vermaelen (1981)): 24.31% for the smallest firms vs. 8.3% for the largest firms. Only very small firms (quintile 1) experience significant negative excess returns in the 40 months before the announcement of -35.2% ($t = -3.33$) and significant positive excess returns of 24.2% ($t = 2.2$) in the 23 months after the expiration. A similar, but less permanent and not statistically significant pattern is observed in the second quintile. When we move to larger firms (quintile 3 thru 5) the pattern changes: large firms don't experience negative but positive excess returns prior to the repurchase. After the expiration of the offer, excess returns are not significantly different from zero.

We conclude that the post-expiration positive drift in security prices is mainly due to the price behavior of very small firms. These firms were doing poorly in the years prior to the repurchase, which signalled a true change in their future performance. The market apparently underestimated the extent of the "good" news so that investors could have earned significant excess returns by buying after the expiration date. Large firms, on the other hand, were doing relatively well prior to the repurchase and may have accumulated excess cash-flows as a result. Hence, the repurchase may have been less motivated by "signalling" explanations and has also generated smaller excess returns. Note that small firms (quintile 1) from the month prior to the repurchase until 24 months afterwards, experienced abnormal returns of 43.02%. The corresponding number for the large firms (quintile 5) is 15.18%.

6 Conclusion

The results of this paper indicate that by executing a simple trading strategy around repurchase tender offers it is possible to generate abnormal returns of more than 9% per week, on average, in the period 1962-1986. Further research is necessary to

⁶We assume that the expiration month is month 1.

analyze (1) why these trading profits are not arbitrated away by risk arbitrageurs and (2) whether developing a model to forecast shareholder and managerial behavior around repurchase tender offers can improve the trading strategy. Even without such a model, we are able to explain 75% of the variability of the trading return on the basis of one variable: the market price before the expiration day divided by the tender price. Because the trading strategy is almost riskless and does not require any sophisticated analysis, its performance is inconsistent with market efficiency and is therefore properly characterized as an “anomaly”.

The results also show that a portfolio of repurchasing companies earns significant positive abnormal returns after the repurchase. This effect is most pronounced for small firms that earn excess returns of 24% in the 23 months after the expiration of the offer. It appears that, on average, the market has underestimated the value of the information signalled thru the repurchase.

Table 1 Descriptive Statistics: Mean values (in%)

	<u>Period</u>		
	1962-1986	1962-1979	1980-1986
# of observations	221	131	90
F_S	17.06	15.68	19.07
F_P	16.41	15.45	17.82
F_P/F_T	86.61	88.67	83.60
<i>PREMIUM</i>	21.79	24.09	18.54
<i>CAR</i>	12.54	14.58	9.78
<i>TOTALR</i>	14.29	16.19	11.52

F_S =fraction of shares sought; F_P =Fraction purchased. F_T =fraction tendered. *PREMIUM*=tender price minus the price 5 days before the announcement divided by the price 5 days before the announcement. *CAR*=cumulative abnormal return from 5 days before the announcement until 10 days after the expiration. *TOTALR*=weighted average abnormal return to all shareholders.

Table 2 Average abnormal Returns (in per cent) from buying T_1 days prior to the expiration date and tendering to the firm, whenever the market price is 3% below the tender price. In oversubscribed offers, shares are sold T_2 days after the expiration day.

Panel A: Total sample(1962 - 1986)
(109 observations)

	<u>Average</u>	<u>t-value</u>	<u>Median</u>	<u>% positive</u>
Trading rule # 1 ($T_1 = 0, T_2 = 2$)	9.61	7.75	5.88	94.5
Trading rule # 2 ($T_1 = -1, T_2 = 2$)	9.46	7.76	5.52	97.2
Trading rule # 3 ($T_1 = -1, T_2 = 4$)	9.26	7.58	5.59	92.7

Panel B: First sub-period (1962 - 1979)
(63 observations)

	<u>Average</u>	<u>t-value</u>	<u>Median</u>	<u>% positive</u>
Trading rule # 1 ($T_1 = 0, T_2 = 2$)	8.42	6.69	5.88	90.6
Trading rule # 2 ($T_1 = -1, T_2 = 2$)	8.01	7.58	5.42	95.4
Trading rule # 3 ($T_1 = -1, T_2 = 4$)	7.82	7.26	5.59	90.7

Panel C: Second sub-period (1980 - 1986)
(46 observations)

	<u>Average</u>	<u>t-value</u>	<u>Median</u>	<u>% positive</u>
Trading rule # 1 ($T_1 = 0, T_2 = 2$)	11.26	4.71	5.48	100.0
Trading rule # 2 ($T_1 = -1, T_2 = 1$)	11.60	4.51	5.71	100.0
Trading rule # 3 ($T_1 = -1, T_2 = 4$)	11.39	4.44	5.33	95.5

Table 3 Average abnormal returns (in per cent) from buying T_1 days prior to the expiration date and tendering to the firm, whenever the market price is 5% below the tender price. In oversubscribed offers, nonrepurchased shares are sold T_2 days after the expiration day.

Panel A: Total sample (1962 - 1986)

(77 observations)

	<u>Average</u>	<u>t-value</u>	<u>Median</u>	<u>% positive</u>
Trading rule # 1 ($T_1 = 0, T_2 = 2$)	12.34	7.38	8.09	86.2
Trading rule # 2 ($T_1 = -1, T_2 = 2$)	11.36	7.1	6.82	96.9
Trading rule # 3 ($T_1 = -1, T_2 = 4$)	11.06	7.0	6.78	95.9

Panel B: First sub-period (1962 - 1979)

(47 observations)

	<u>Average</u>	<u>t-value</u>	<u>Median</u>	<u>% positive</u>
Trading rule # 1 ($T_1 = 0, T_2 = 2$)	10.46	6.53	8.09	84.1
Trading rule # 2 ($T_1 = -1, T_2 = 2$)	9.16	6.93	6.45	95.2
Trading rule # 3 ($T_1 = -1, T_2 = 4$)	8.86	6.56	6.35	93.6

Panel C: Second sub-period (1980 - 1986)

(30 observations)

	<u>Average</u>	<u>t-value</u>	<u>Median</u>	<u>% positive</u>
Trading rule # 1 ($T_1 = 0, T_2 = 2$)	15.29	4.42	7.39	97.9
Trading rule # 2 ($T_1 = -1, T_2 = 1$)	11.62	4.16	7.77	100
Trading rule # 3 ($T_1 = -1, T_2 = 4$)	14.71	4.08	7.45	100

Table 4 Results from the regression $XRET_i = \alpha + \beta \cdot \frac{P_T - P_B}{P_B}_i + \varepsilon_i$ where P_T = tender price, P_B is the price T_1 days before the expiration and $XRET$ is the abnormal return from buying shares T_1 days before the expiration day and selling them T_2 days after. (t - statistics in parentheses). There are 193 observations.

T_1, T_2	α	β	$\bar{R}^2$
$T_1 = 0 \ T_2 = 2$	-0.008 (-1.81)	0.860 (24.82)	0.76
$T_1 = -1 \ T_2 = 2$	-0.008 (-1.79)	0.847 (25.42)	0.80
$T_1 = -1 \ T_2 = 4$	-0.011 (-2.28)	0.845 (23.84)	0.78

Table 5A Average and median ratio of trading volume relative to normal trading volume around the announcement of the offer.

Panel A: Total period (171 observations)

day	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1
Mean ¹	1.30	1.27	1.55**	1.19	1.41**	1.26	1.53**	1.86**	1.56**	6.78*
Median	1.77	0.90	0.97	0.85	0.83	0.77	0.92	0.98	0.92	2.87
% > 1	38	46	49	40	42	40	49	49	45	76

day	0	1	2	3	4	5	6	7	8	9	10
Mean	8.89*	5.77*	3.80*	3.71*	4.52*	3.19*	3.04*	3.58*	3.79*	3.76*	3.19*
Median	4.63	3.03	2.21	2.06	1.93	1.62	1.56	1.32	1.51	1.50	1.25
% > 1	83	79	74	71	68	65	65	64	61	63	58

Panel B: First sub-period (1962-1979, 86 observations)

day	-20	-19	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1
Mean	1.11	1.41	1.13	1.20	1.56	1.17	1.41	1.31	1.31	1.51*	1.37**	8.26*
Median	0.86	0.83	0.82	0.92	0.94	0.82	0.71	0.82	0.82	0.94	0.87	3.23
% > 1	44	41	36	46	46	38	39	42	47	48	42	77

day	0	1	2	3	4	5	6	7	8	9	10
Mean	10.04*	5.82*	4.51*	4.30*	5.71*	3.55*	3.00*	3.68*	4.40*	3.11*	2.62*
Median	4.65	2.84	2.19	2.04	1.89	1.60	1.55	1.20	1.32	1.50	1.08
% > 1	81	74	74	69	69	63	66	63	56	63	52

Panel C: Second sub-period (1980-1986, 85 observations)

day	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1
Mean	1.47	1.33	1.55	1.20	1.42	1.20	1.75**	2.22**	1.77**	5.27*
Median	0.75	0.87	1.00	0.85	0.92	0.68	1.03	1.00	0.97	2.67
% > 1	39	45	52	42	45	39	51	49	49	75

day	0	1	2	3	4	5	6	7	8	9	10
Mean	7.72*	5.72*	3.07*	3.11*	3.27*	2.81*	3.08*	3.48*	3.17*	4.45*	3.76*
Median	4.48	3.09	2.31	2.17	2.03	1.62	1.58	1.74	1.55	1.49	1.45
% > 1	84	83	75	74	69	66	64	65	65	64	64

¹ Under the null hypothesis, the mean ratio is one. Note that the median ratio for normal periods is less than one because the distribution of the ratio is skewed to the right.

* significantly different from one at the 1 percent level (two-tailed test)

** significantly different from one at the 5 percent level (two-tailed test)

Table 5B Average and median ratio of trading volume relative to normal trading volume 10 days before the expiration day until 10 days after.

Panel A: Total period (140 observations)

day	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1
Mean ¹	5.40*	4.22*	5.21*	3.53*	2.72*	3.40*	3.18*	2.32*	3.38*	2.50*
Median	1.67	1.48	1.52	1.36	1.54	1.36	1.27	1.13	1.19	1.03
%>1	67	66	66	64	58	61	58	55	52	51

	0	1	2	3	4	5	6	7	8	9	10
Mean	2.65*	2.06*	1.42*	1.06	1.36	1.22	0.92	1.07	0.98	1.02	1.36
Median	1.38	1.15	0.89	0.61	0.65	0.68	0.53	0.61	0.55	0.58	0.67
%>1	58	58	48	32	29	33	28	33	33	28	30

Panel B: First sub-period (1962 - 1979, 77 observations)

day	-10	-9	-8	-7	-6	-5	-4	-3	-2	1
Mean	5.58*	5.21*	6.53*	3.34**	2.55**	3.01**	3.37**	2.21**	2.50**	2.32**
Median	1.73	1.45	1.42	1.31	1.22	1.07	1.04	1.13	1.39	1.09
%>1	68	64	67	61	54	53	51	52	59	53

	0	1	2	3	4	5	6	7	8	9	10
Mean	2.21**	2.20**	1.49	1.24	1.18	1.29	0.75	1.10	0.84	0.96	1.15
Median	1.38	1.12	0.76	0.58	0.55	0.58	0.40	0.56	0.42	0.56	0.56
%>1	59	59	45	35	28	28	21	28	23	19	25

Panel C: Second subperiod (1980 - 1986, 63 observations)

day	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1
Mean	5.18*	3.01*	3.65*	3.74*	2.92*	3.85*	2.95*	2.45*	4.47*	2.73*
Median	1.62	1.52	1.71	1.75	1.67	1.48	1.65	1.22	0.86	0.96
%>1	67	68	66	67	64	71	65	58	45	48

day	0	1	2	3	4	5	6	7	8	9	10
Mean	3.19*	1.91*	1.34**	0.85	1.58	1.13	1.13	1.03	1.13	1.09	1.62
Median	1.25	1.27	1.04	0.61	0.68	0.71	0.74	0.67	0.94	0.68	0.79
%>1	58	57	53	28	30	38	36	38	43	38	37

¹ Under the null hypothesis, the mean ratio is one. Note that the median ratio for normal periods is less than one because the distribution of the ratio is skewed to the right.

* significantly different from one at the 1 percent level (two-tailed test)

** significantly different from one at the 5 percent level (two-tailed test)

Table 6 Average abnormal returns (in percent) from buying 6 days prior to the expiration date and tendering to the firm, whenever the market price is $x\%$ below the tender price. In oversubscribed offers, the nonrepurchased shares are sold 12 trading days after the expiration day.

PANEL A $x = 3\%$

	<u>Average</u>	<u>t - value</u>	<u>Median</u>	<u>% positive</u>
<u>1962-1986</u>				
(110 observations)	6.18	4.95	4.64	89.1
<u>1962-1979</u>				
(62 observations)	5.59	4.57	5.38	88.7
<u>1980-1986</u>				
(48 observations)	6.94	2.90	3.15	89.6

PANEL B $x = 5\%$

	<u>Average</u>	<u>t - value</u>	<u>Median</u>	<u>% positive</u>
<u>1962-1986</u>				
(67 observations)	9.32	4.97	5.67	91.0
<u>1962-1979</u>				
(39 observations)	7.36	4.31	7.20	89.7
<u>1980-1986</u>				
(28 observations)	12.06	3.18	5.20	92.9

Table 7 The average (AR) and cumulative average (CAR) excess returns (in %) from 2 days before the expiration date until six days after the expiration date for 2 subsamples. Panel A is based on all 121 oversubscribed offers. Panel B is based on a subsample of 52 oversubscribed offers where the management purchased at least 20% more shares than it had to. (t - values in parentheses).

Panel A: All oversubscribed

day	-2	-1	0	1	2	3	4	5	6
1962-1986			121	observations					
AR	0.03	-0.25	-0.11	-1.21	0.30	-0.18	-0.05	0.07	-0.12
(t)	(0.17)	(-1.48)	(-0.58)	(-2.80)	(0.76)	(-0.68)	(-0.23)	(0.38)	(-0.37)
CAR	0.03	-0.21	-0.32	-1.53	-1.24	-1.41	-1.46	-1.39	-1.51
1962-1979			71	observations					
AR	0.14	-0.36	0.12	-1.76	0.53	-0.17	-0.01	0.22	0.04
(t)	(0.54)	(-1.45)	(0.53)	(-2.58)	(0.83)	(-0.46)	(-0.03)	(0.88)	(0.14)
CAR	0.14	-0.22	-0.10	-1.86	-1.33	-1.50	-1.51	-1.29	-1.25
1980-1986			50	observations					
AR	-0.12	-0.08	-0.43	-0.43	-0.04	-0.19	-0.11	-0.15	-0.34
(t)	(-0.52)	(-0.43)	(-1.48)	(-1.12)	(-0.13)	(-0.53)	(-0.33)	(-0.60)	(-1.34)
CAR	-0.12	-0.21	-0.64	-1.07	-1.11	-1.29	-1.40	-1.55	-1.88

Panel B: cases where $(F_p - F^*)/F^* > 0.2$

day	-2	-1	0	1	2	3	4	5	6
1962-1986			52	observations					
AR	0.13	-0.36	-0.38	-0.70	1.23	0.23	0	0.04	0.06
(t)	(0.48)	(-1.63)	(-1.00)	(-0.88)	(1.67)	(0.58)	(0.00)	(0.10)	(0.21)
CAR	0.13	-0.23	-0.62	-1.32	-0.08	0.15	0.14	0.18	0.24
1962-1979			36	observations					
AR	0.13	-0.65	-0.21	-0.74	1.40	0.41	0.18	0.16	0.03
(t)	0.34	(-2.24)	(-0.61)	(-0.66)	(1.35)	(0.73)	(0.38)	(0.44)	(0.05)
CAR	0.13	-0.52	-0.73	-1.47	-0.06	0.35	0.52	0.68	0.71
1980-1986			16	observations					
AR	0.12	0.30	-0.79	-0.61	0.85	-0.18	-0.40	-0.23	0.13
(t)	(0.36)	(1.15)	(-2.13)	(-0.64)	(1.70)	(0.26)	(-0.57)	(0.35)	(0.34)
CAR	0.12	0.42	-0.36	-0.97	-0.12	-0.31	-0.77	-0.94	-0.80

Table 8 Regression of F_p/F_T against P_B/P_T , for 120 oversubscribed offers (t - values in parentheses)

$\frac{F_p}{F_T}$	$= 1.14 - 0.45 \frac{P_B}{P_T}$	$\overline{R}^2 = 1.26\%$
	(4.87) (-1.59)	

Table 9 Cumulative average abnormal returns (in %) from the announcement month until 24 months after the repurchase, for 4 benchmarks: the value-weighted market index (VW), the equally-weighted market index (EW), portfolios with the same size (SIZE) and portfolios with the same size and beta. (SIZE & BETA)

Total period (1982-1986)				
<u>MONTH</u>	<u>VW</u>	<u>EW</u>	<u>SIZE</u>	<u>SIZE & BETA</u>
0	13.82	13.57	13.62	13.89
1	12.58	11.74	11.41	11.79
2	13.16	11.83	11.49	12.12
3	13.15	11.28	10.76	11.60
4	14.69	12.37	11.71	12.57
5	15.14	12.72	12.25	13.25
6	14.79	12.06	11.48	12.45
7	16.31	13.12	11.99	13.37
8	17.02	13.25	11.93	13.43
9	18.94	14.46	13.01	14.62
10	19.85	14.84	12.99	14.19
11	21.99	16.69	14.80	15.99
12	21.75	16.24	14.51	15.88
16	25.87	18.20	15.53	16.95
20	32.98	22.88	18.90	20.49
24	36.27	24.17	20.06	20.88

Table 10 Cumulative average abnormal returns (in %) from the announcement month until 24 months after the repurchase, for 4 benchmarks: the value-weighted market index (VW), the equally-weighted market index (EW), portfolios with the same size (SIZE) and portfolios with the same size and beta. (SIZE & BETA)

First subperiod (1962-1979)

<u>MONTH</u>	<u>VW</u>	<u>EW</u>	<u>SIZE</u>	<u>SIZE & BETA</u>
0	16.04	15.75	15.78	16.03
1	14.55	13.66	14.77	13.32
2	14.78	13.31	13.15	13.58
3	15.01	12.94	12.43	13.15
4	16.84	14.24	13.50	14.51
5	17.29	14.61	14.21	15.26
6	16.97	14.02	13.57	14.47
7	19.75	16.18	15.00	16.27
8	19.94	15.72	14.21	15.48
9	23.02	17.82	15.98	17.32
10	23.61	17.92	15.58	16.52
11	25.65	19.53	17.13	18.04
12	25.41	19.26	16.89	17.99
16	31.68	22.57	18.79	19.89
20	40.42	27.90	22.07	22.87
24	45.64	30.11	23.59	23.60

Table 11 Cumulative average abnormal returns (in %) from the announcement month until 24 months after the repurchase, for 4 benchmarks: the value-weighted market index (VW), the equally-weighted market index (EW), portfolios with the same size (SIZE) and portfolios with the same size and beta. (SIZE & BETA)

Second subperiod (1980-1986)

<u>MONTH</u>	<u>VW</u>	<u>EW</u>	<u>SIZE</u>	<u>SIZE & BETA</u>
0	10.54	10.34	10.44	10.72
1	9.66	8.90	8.84	9.54
2	10.77	9.64	9.29	9.96
3	10.39	8.81	8.45	9.32
4	11.45	9.59	9.06	9.68
5	11.95	9.91	9.33	10.24
6	11.55	9.14	8.37	9.44
7	11.11	8.52	7.46	9.02
8	12.63	9.55	8.52	10.38
9	12.76	10.41	8.54	10.57
10	14.12	10.21	9.11	10.72
11	16.47	12.43	11.34	12.95
12	16.22	11.78	10.94	12.73
16	17.00	11.57	10.63	12.52
20	21.35	15.14	14.65	17.01
24	21.03	14.69	14.53	16.73

Table 12 Size distribution of the sample. Number of firms in each size decile

<u>Decile</u>	<u>First period</u> (1962-1979)	<u>Second period</u> (1980-1986)	<u>Total</u> (1962-1986)
1 (smallest)	8	6	14
2	17	11	28
3	27	12	39
4	18	7	25
5	17	8	25
6	17	9	26
7	18	7	25
8	10	12	22
9	16	14	30
10(largest)	7	20	27
Total	153	105	258

Table 13 Cumulative average abnormal return (adjusted for beta and size) from month t_1 until month t_2 , $CAR(t_1, t_2)$, for 5 size based subgroups (in %). T-statistics in parenthesis.

<u>Size quintile</u>	<u>CAR (-40, -1)</u>	<u>CAR (0,0)</u>	<u>CAR (2,24)</u>
1 (smallest)	-35.2 (-3.33)	24.3 (9.01)	24.2 (2.02)
2	-10.6 (-1.13)	16.8 (8.15)	13.9 (1.45)
3	2.5 (0.1)	11.2 (6.13)	0.8 (0.04)
4	24.03 (1.62)	10.5 (4.72)	-0.1 (-0.06)
5 (largest)	9.9 (0.9)	8.3 (5.10)	7.40 (1.78)

REFERENCES

- Bagnoli, Mark, Roger Gordon and Barton L. Lipman, 1987, Takeover bids, defensive stock repurchases and the efficient allocation of corporate control, manuscript.
- Bagwell, C., 1988, Share repurchase and takeover difference, manuscript, Northwestern University.
- Banz, Rolf, 1981, The relationship between return and market value of common stock, Journal of Financial Economics, 9, 3-18.
- Brown, S. J. and J. B. Warner, 1985, Using daily stock returns: the case of event studies, Journal of Financial Economics 14, 3-31.
- Choi, Desong, 1987, Informational asymmetry, signalling and insider trading: The case of stock repurchase tender offer, manuscript, SUNY.
- Dann, Larry, 1981, Common stock repurchases: an analysis of returns to bondholders and stockholders, Journal of Financial Economics, 9, 115-138.
- Dimson, Elroy and Paul Marsh, 1986, Event study methodologies and the size effect: the case of UK press recommendations, Journal of Financial Economics. 17, 113-143.
- Jensen, Michael, 1986a, The takeover controversy, Midland Corporate Finance Journal.
- Jensen, Michael, 1986b, Don't freeze the arbs out, Wall Street Journal, Dec. 3, p26.
- Lacker, David and Thomas Lys, 1987, An empirical analysis of the incentives to engage in costly information acquisition: the case of risk arbitrage, Journal of Financial Economics, 18, 11-126.
- Lakonishok, Josef and Seymour Smidt, 1984, Volume and turn of the year behavior, Journal of Financial Economics, 13, 435-457.
- Loomis, Carol, 1986, Beating the market by the buying back stock, Fortune Magazine, April 29.
- Masulis, Ron, 1980, Stock repurchase by tender offer: an analysis of common stock price changes, Journal of Finance, 35, 305-318.
- Poulsen, Arnette and Gregg Jarrell, 1988, Stock trading before the announcement of tender offers: insider trading or market anticipation?, manuscript.
- Rosenfeld, A, Repurchase offers: information adjusted premiums and shareholder's response, working paper, Krannat Graduate School of Management, Purdue University.
- Vermaelen, Theo, 1981, Common stock repurchases and market signalling, Journal of Financial Economics, 9, 139-183.

- Vermaelen, Theo, 1984, Repurchase tender offers, signalling and managerial incentives, Journal of Financial and Quantitative Analysis, 19, 163-181.
- Weinstein, 1984, Arbitrage, in the Complete Guide to Investment Opportunities ed. by Marshall Blume and Jack Freidman, p 1- 17.

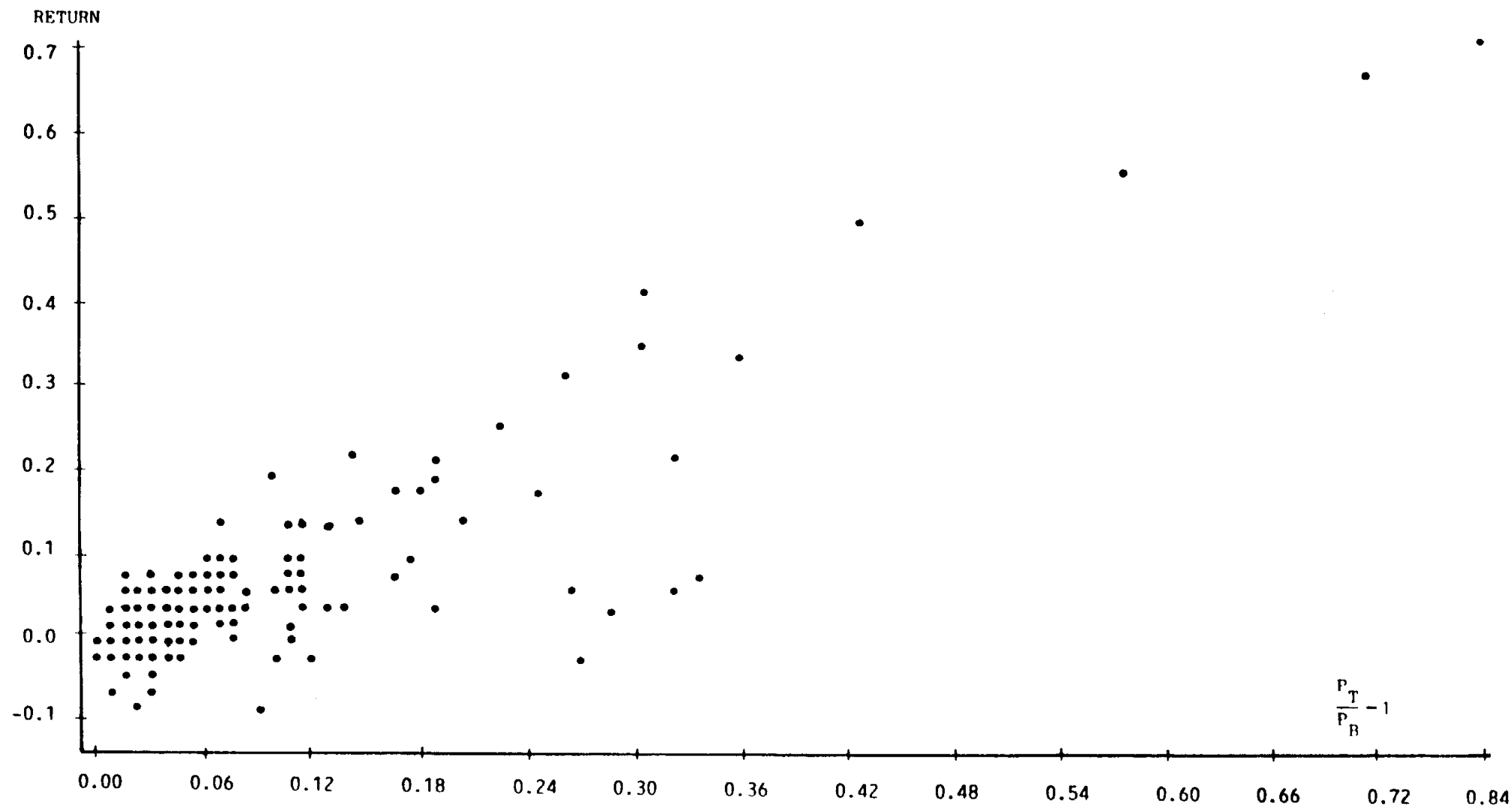


Fig 1 Relationship between trading return and $(\frac{P_T}{P_B} - 1)$

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